COMPUTATIONAL EVIDENCE DOSSIER

This document describes the evidence produced by a run, what can be reproduced, and what was not evaluated. It does not rate the molecule or claim that it is a drug, clinical candidate, or safe compound.

Field	Content	Status
Case	QA cierre 8HFI	RECORDED
Molecule	e5655ee8-fa0b-4c1a-aebe-995548b0281b	RECORDED
Receptor	8HFI · CRYO-EM STRUCTURE OF HUMAN NOREPINEPHRINE TRANSPORTER NET IN THE	RECORDED
Receptor support	Structurally curated; calibration and a family pipeline are missing — the receptor's Spearman ρ and calibrated stacking for its family are missing	RECORDED
Catalog section	Demo Receptor prepared automatically from the PDB, without pose verification or calibration. It serves to explore the application, not to decide what to synthesize.	REVIEW
Run	27c62c96-a890-47f6-b0c1-095dc6801edb	RECORDED
Run date	2026-10-10T00:52:18.887044	RECORDED
Dossier schema	v1	RECORDED

1. Case question and purpose

Field	Content	Status
Study type	explore-hypothesis	RECORDED
What is this case trying to answer?	NOT DEFINED The case did not declare it.	NOT DEFINED
What decision is intended?	NOT DEFINED The case did not declare it.	NOT DEFINED
What supports the choice of system?	NOT DEFINED The case did not declare it.	NOT DEFINED
What controls or references are available?	NOT DEFINED The case did not declare it.	NOT DEFINED

2. Summary of what ran

Field	Content	Status
Execution status (technical)	completed	RECORDED
Scientific disposition of the evidence	REVIEW The run produced useful evidence, but it contains caveats that should be reviewed before proceeding.	REVIEW
Human review	NOT EVALUATED No one has reviewed or acknowledged this run's checks.	NOT EVALUATED

Technical status, scientific disposition, and human review are three separate things. A process finishing does not validate its result, and the absence of alerts does not mean anyone reviewed it.

3. Inputs and provenance

Field	Content	Status
Declared receptor	8HFI · chain A	RECORDED
Docked receptor (prepared unit)	8HFI · cadenas A (measured from the hashed PDBQT bytes)	RECORDED
Receptor source	curado	RECORDED
Input ligand	c1ccccc1CNCC(F)(F)F	RECORDED
Canonical ligand (the one docked)	FC(F)(F)C[NH2+]Cc1ccccc1	RECORDED
Effective search box	center (100.281, 100.843, 101.442) · size (30.0, 30.0, 30.0) Å	RECORDED
Reference residues	A:ALA145, A:ASP75, A:GLY149, A:GLY320, A:GLY423, A:LEU319, A:PHE317, A:PHE323, A:PHE72, A:SER318, A:SER419, A:SER420, A:TYR152, A:VAL148, A:VAL325	RECORDED
Input fingerprint (preflight)	sha256:f5434a08d18f4ae81fa2791292d093caf2769638797d0149fafa660121e34e74	RECORDED
Relationship of the run with current inputs	This run used the inputs described in this dossier.	RECORDED
Box declared by the case	center (100.281, 100.843, 101.442) · size (30.0, 30.0, 30.0) Å Value declared by the case; it does not replace the box used.	RECORDED

Field	Content	Status
Ligand hash (SMILES)	e3dc5e9f3439b3ff841416be2b6dc76e8b3b1538e77c1add165bcfc75d8259b3	RECORDED
Receptor reference	runs/receptors/30e6695403f7b072cbe436daefa3479f9c8e33ee670590e5f04e01d3a7ff0e60.pdbqt	RECORDED
Configuration hash	NOT AVAILABLE No sealed configuration.	NOT AVAILABLE
Pose hash	c49f7d5ef53c292a65f29da5923838ef1e5d0f2c4a042e3459aa0c9fae179170	RECORDED

4. System preparation and assumptions

Field	Content	Status
Declared heteroatom policy	docking_vina	RECORDED
Waters, metals, and cofactors	HETATM records removed from the assembly: waters 39, metals 0, organic cofactors 0, other species 21. Within 6 Å of the box: removed 32 water(s), CL A:702 (0.0 Å); retained none. The crystal ligand (DSM A:701) was removed as part of the run. Preparation removed metals, cofactors, or other species near the site: the docked receptor does not contain them. Decide whether they were part of the cavity.	REVIEW
Ligand stereochemistry	FC(F)(F)C[NH2+]Cc1ccccc1 Isomeric SMILES derived from the saved 3D conformer.	RECORDED
Preflight warnings	REVIEW Scientific warnings recorded before execution.	REVIEW
Unassessed checks in the previous check	ABSTENTION The preflight could not execute these controls. They were not evaluated: that does not mean they passed them.	ABSTENTION

Human decisions: NOT EVALUATED — no one has reviewed or acknowledged this run's checks.

5. Protocol and run environment

Field	Content	Status
Docking engine	vina	RECORDED
Engine version	1.2.7	RECORDED
MolDesign version	1.0.1	RECORDED
Code commit	NOT AVAILABLE This run did not declare a verifiable commit (bundle or modified tree).	NOT AVAILABLE
CPUs requested for Vina	4	RECORDED
Prepared receptor SHA-256	30e6695403f7b072cbe436daefa3479f9c8e33ee670590e5f04e01d3a7ff0e60	RECORDED
Input ligand SDF SHA-256	680cf104e8d0268e11eaef32c731cb79f0331119ebc5f49b3c5ad86ffe6ed4e5	RECORDED
PDBQT ligand SHA-256	8efe1b0ab321873f882f4737d1eb3dff756741c479e747ca24cd42b4e6473f40	RECORDED
Pose SDF SHA-256	6131119e2272eb427c1eb97c7f81b64a66596900c1f2a71c9b2a825b7b07a79e	RECORDED
Log SHA-256	e6a12bf94ad5a88297ee6ff5e8238302681241ecb8a7dab705b1584dabcd391b	RECORDED
Receptor preparation recipe	cde3e715fd357bd60903e848e66001d195d8d31699beef3ee666f26c400d0efc	RECORDED
Preparation versions	{"dimorphite_dl": "2.0.2", "meeko": "0.7.1", "rdkit": "2025.9.6", "vina": "1.2.7"}	RECORDED
Random seed	42	RECORDED
Parser source	sdf	RECORDED
Rescoring model	universal	RECORDED
Declared PRO settings	{"docking_engine": "vina", "enabled_stages": ["validation", "properties", "sa_filter", "conformer", "docking", "xgb", "clgnn"], "gnn_precision": "fp32", "pro_anti_targets": [], "pro_mmgbsa": false, "pro_mmgbsa_steps": 1000, "pro_parallel_docks": 2, "pro_selectivity": true, "pro_workers": 4, "stage_params": {"conformer": {"conformers": 1, "ph": 7.4}, "docking": {"exhaustiveness": 8, "num_poses": 9}, "properties": {"run_admet_ai": false}}}	RECORDED
Exhaustiveness	8	RECORDED
Poses requested	9	RECORDED
Fallback reason	Familia 'transporter' sin modelo entrenado disponible — modelo universal usado.	RECORDED
Declared error	NOT AVAILABLE This run did not record errors.	NOT AVAILABLE
Exhaustiveness declared by the case	8 Value declared by the case; execution is documented separately.	RECORDED
Poses declared by the case	9 Value declared by the case; execution is documented separately.	RECORDED

6. Structural evidence and docking

Rank	Affinity (kcal/mol)	RMSD lower bound	RMSD upper bound
1	-6.95	0.00	0.00
2	-6.76	3.22	4.43
3	-6.64	3.53	5.23
4	-6.54	3.63	5.29
5	-6.41	4.68	6.18
6	-6.36	4.38	5.61
7	-6.30	4.35	5.43
8	-6.26	3.90	5.42
9	-6.25	4.38	5.66

Affinities are ranking signals within this protocol, not measurements of free energy. The pose with the best energy is not necessarily the correct pose: without a passed geometric check, there is no structural evidence.

7. Pose generation

Field	Content	Status
Poses generated	9	RECORDED
Poses requested from the engine	9	RECORDED
Exhaustiveness	8	RECORDED
Random seed	42	RECORDED
Reference residues	15 selected out of 15 in the catalog: A:ALA145, A:ASP75, A:GLY149, A:GLY320, A:GLY423, A:LEU319, A:PHE317, A:PHE323, A:PHE72, A:SER318, A:SER419, A:SER420 … and 3 more	RECORDED
Engine and version	vina · 1.2.7	RECORDED
Conformer generation	Single conformer (ETKDG + MMFF94)	RECORDED
Best observed Vina affinity	-6.95 kcal/mol	RECORDED
Physical validation coverage	9/9 poses with a verdict	RECORDED
Observation · AutoDock Vina affinity	-6.95 kcal/mol (Vina score, untransformed)	RECORDED
Interpretation · regression XGBoost (pKi)	pKi 3.56 · equivalent to -4.84 kcal/mol · within the applicability domain It is reported TOGETHER the affinity of Vina, not in its place: they are two quantities with different error and different domain of validity.	RECORDED
Rescoring model and structural family	universal · transporter family	RECORDED
Effective M4 stacking weights	vina=1.0000 | xgb=0.0000 | gnn=0.0000 | clgnn=0.0000 The normalized weights actually used in this run were persisted.	RECORDED
Engine used	vina	RECORDED
RDKit version for canonicalization	2025.09.6	RECORDED
Identity v2 SHA-256	ac25507fefae3cc4a7fbed2f0246e315d0e97b1880afc11289dbe77b3ea93dcb	RECORDED
ligand protonation pH	7.4	RECORDED
Docked species	REVIEW Microstate closest to the one predicted at pH 7.4, out of 2 enumerated. The selection is reproducible, but the class-based pKa table does not resolve the population of this class. Under review because of protonated amine with an electron-withdrawing group one or two bonds from the N (CF3 or CHF2 at β, nitrile at α or β): the group lowers the class pKa the table assigns (9.8) by several units; 2,2,2-trifluoroethylamine has pKa ≈5.7 and 3-aminopropionitrile ≈7.8, so the charge at pH 7.4 is an open decision. Review the species before a free-energy calculation.	REVIEW
Ligand tautomer	The RDKit canonical tautomer was docked; protonation follows the declared pH. One enumerable tautomer	RECORDED
Ligand efficiency (LE)	0.535 kcal/mol per heavy atom (13 atoms). Hopkins et al. 2004. It depends on size by construction: measured across this project's cohort (17,431 molecules), its correlation with atom count is as strong as that of the raw score, with the opposite sign.	RECORDED
SILE	3.221 = |affinity| / HA^0.3. Nissink, J. Chem. Inf. Model. 49 (2009) 1617. The paper does not publish a threshold, so none is used here.	RECORDED
Fit Quality (Vina proxy)	0.675 = LE / LE_scale(HA), where LE_scale = 0.0715 + 7.5328/HA + 25.7079/HA² − 361.4722/HA³ (Reynolds, Tounge, Bembenek, J. Med. Chem. 51 (2008) 2432, Eq. 1–2). LE_scale used: 0.5808 in pKi units. PROXY: the scale was fitted against experimental Ki, not docking scores; the conversion uses 1.37 kcal/mol per log unit. FQ ≈ 1 means proximity to the efficient frontier of the Reynolds set for that size—it is not a probability of binding or pose quality. The scale was capped at HA=15: the Reynolds fit covers 15–50 heavy atoms, and this molecule falls outside that range, so the value is a bound, not an interpolation.	RECORDED

Parameters that the run did not persist appear as NOT EVALUATED with a reason. They are not replaced with values from the case configuration: that would describe settings no one saved with this result.

8. Pose selection

Field	Content	Status
Selection stage status	SELECTOR ABSTENIDO The margin did not reach the threshold: the selector does NOT recommend a pose.	ABSTENTION
Product primary pose (Vina top-1)	#1	RECORDED
Pose recommended by the selector	NOT AVAILABLE The selector did not recommend a pose.	NOT AVAILABLE
Pose the selector would have recommended (abstention)	#1	RECORDED
Selector margin and threshold	0.081710 (threshold 0.097663)	RECORDED
Declared reference	Vina top-1 as FALLBACK: this is not a selector hit When there is no recommendation, Vina top-1 is the default reference.	REVIEW
Abstention reason	MARGEN_BAJO_UMBRAL	RECORDED
Selector model	v0.6 XGBRanker rank:pairwise (selector oficial de Ruta C)	RECORDED
Model SHA-256	9827ddb94c5ded5b2ef1dda1250606d73f73163becc310641d5bd493196e5d31	RECORDED
Model metadata SHA-256	9ac4e379f9c54aef24acc72fe7e61ffff751a70bf80b751f610bf93f34e4fc96	RECORDED
Physically Valid Alternatives	#1, #2, #3, #4, #5, #6, #7, #8, #9 They are listed for comparison. NONE is selected automatically.	RECORDED

The selector RECOMMENDS; it does not replace. The product's primary pose remains Vina top-1, which is always retained and declared. If the selector abstains, is missing, or fails, the reference is Vina top-1 as a FALLBACK—not as a selector success. The confidence margin is the distance between the first and second pose according to the model; it is not a probability or a score.

9. Geometric and physical validation

Field	Content	Status
Stage verdict	CHECKS PASSED 9 of 9 poses received a verdict, and none failed the chemistry or geometry checks. Engine: posebusters:0.6.5:dock. Coverage: 9/9 poses.	PASSED
Validation engine	posebusters:0.6.5:dock	RECORDED
Coverage	9 of 9 poses	RECORDED
Reason code	NOT AVAILABLE The stage did not record a reason code.	NOT AVAILABLE
Physical status of the recommended pose	NOT AVAILABLE There is no recommended pose, or its physical status was not measured.	NOT AVAILABLE
Physical status of the pose the selector would have recommended	passed	RECORDED

Verdict by pose

Pose	Vina affinity (kcal/mol)	Physical checks	Selector	Role	Failed checks
#1	-6.95	PASSED	0.3240	Vina top-1 · Valid alternative	—
#2	-6.76	PASSED	0.1456	Valid alternative	—
#3	-6.64	PASSED	0.0974	Valid alternative	—
#4	-6.54	PASSED	0.0045	Valid alternative	—
#5	-6.41	PASSED	0.2422	Valid alternative	—
#6	-6.36	PASSED	0.1990	Valid alternative	—
#7	-6.30	PASSED	-0.1868	Valid alternative	—
#8	-6.26	PASSED	0.0277	Valid alternative	—
#9	-6.25	PASSED	-0.1366	Valid alternative	—

Only an explicit PASSED CHECK allows a pose to be called valid. REVIEW describes a check suite that did not run in full, and NOT EVALUATED describes a validator that did not run: both describe what happened to the check, never to the molecule, and neither is negative

evidence about the compound.

10. Physical and geometric checks

Check	Observation and protocol	Status
Physical and geometric validity of the poses VALIDEZ_FISICA_POSES	9 of 9 poses received a verdict, and none failed the chemistry or geometry checks. Engine: posebusters:0.6.5:dock. Coverage: 9/9 poses. Protocolo: Pose reconstruction from PDBQT using the SMILES template, followed by geometric validation. The representation used for validation is NOT the crystal structure: hydrogens are added during reconstruction. Procedencia: eval_result.structural_evidence	PASSED
Energy separation between poses SEPARACION_ENTRE_POSES	9 poses are within 1 kcal/mol; the pose 1–2 difference is 0.19 kcal/mol. No replicates or ensemble were recorded. Protocolo: Affinity difference between poses 1 and 2, and the count within ≤1 kcal/mol. Procedencia: eval_result.docking_poses	REVIEW
Technical reproducibility trace TRAZA_REPRODUCIBILIDAD	3/3 traces: engine version, seed, and parser. Protocolo: Presence of the three fields needed to repeat the run. Procedencia: eval_result	RECORDED

The absence of alerts does not establish validity: a check that did not run appears as NOT EVALUATED and supports no conclusion.

11. Evidence by dimension

Check	Observation and protocol	Status
System preparation SYSTEM	The inventory records the prepared receptor, grid, and structural references. Procedencia: build_evidence_summary	RECORDED
Docking signal DOCKING	9 serialized poses; best raw affinity -6.95 kcal/mol · vina 1.2.7. Procedencia: build_evidence_summary	RECORDED
Pose sampling SAMPLING	9 poses are within 1 kcal/mol; the pose 1–2 difference is 0.19 kcal/mol. No replicates or ensemble were recorded. Procedencia: build_evidence_summary	REVIEW
Technical reproducibility REPRODUCIBILITY	Minimum traces 3/3 (engine version, seed, and parser); reconstruction items 7/7. Reconstruction by a third party: not demonstrated. Procedencia: build_evidence_summary	RECORDED
Physical validity of poses PHYSICAL	9 of 9 poses received a verdict, and none failed the chemistry or geometry checks. Engine: posebusters:0.6.5:dock. Coverage: 9/9 poses. Procedencia: build_evidence_summary	RECORDED
Model applicability domain MODEL	universal model; fallback: Familia 'transporter' sin modelo entrenado disponible — modelo universal usado. The docking engine and its version are listed in the reproducibility trace, not here; they are distinct identities. Procedencia: build_evidence_summary	RECORDED
Physicochemical descriptors PHYSICOCHEMICAL	Values and rules are reported separately; they do not become a global rating. Procedencia: build_evidence_summary	RECORDED
Selectivity evidence SELECTIVITY	Computational comparison relating to the protocol; it does not demonstrate experimental selectivity. Procedencia: build_evidence_summary	REVIEW

12. Declared assumptions

• Affinities are interpreted as ranking signals within the protocol, not as experimental free-energy measurements.

• The best raw pose is identified by serialized affinity order; that order does not demonstrate geometric validity.

• No preparation, protonation, or minimization steps are attributed to the run unless it recorded them.

• Desolvated preparation: all crystallographic waters were removed from the receptor. If this target requires conserved bridging waters for molecular recognition, this run's contacts and affinities do not include them.

13. Uncertainties and limitations

• This receptor has no measured Spearman ρ, and its structural family has no calibrated stacking, so it runs under the shared conservative alpha contract. There is no evidence that the ranking correlates with actual affinity here. Treat the result as a hypothesis to test, not as a measurement.

• 9 poses are within 1 kcal/mol; selecting a single pose is ambiguous.

• The pose selector abstained: it did not recommend a pose. The fallback reference is Vina top-1, not a successful selector prediction.

• Meeko-assigned histidine templates: A:178=HIE, A:220=HIE, A:222=HIE, A:228=HIE, A:280=HIE, A:296=HIE, A:370=HIE, A:372=HIE, A:441=HIE, A:598=HIE, A:599=HIE, A:613=HIE. The assignment is template-based and does not validate the experimental microstate or a specific pH.

• Desolvated preparation: 23 crystallographic waters were removed from the docking box; 10 had at least 3 contacts within 3.5 Å of the receptor (the best-coordinated water had 6). If any of these waters participates in recognition by this target—such as the catalytic water in HIV-1 protease or the hinge network in some kinases—this run's contacts do not include it.

• Ligand efficiency LE=0.53. With 13 heavy atoms, the 0.45 threshold is crossed by any affinity better than -5.9 kcal/mol, so exceeding it does not distinguish good binding from mediocre binding. In this mass range (< 300 Da, Rule of Three), efficiency per atom is high by construction: it can compare molecules of the same size, but cannot serve as a verdict.

• The docked species differs from the input: FC(F)(F)CNCc1ccccc1 → FC(F)(F)C[NH2+]Cc1ccccc1. Unevaluated alternatives remain a source of chemical uncertainty.

14. Supported interpretation and next steps

RESOLVE CAVEATS BEFORE COMPARING

Document the reservations of preparation, domain or reproducibility before comparing this run with others.

None of these actions claims that the compound is a drug, clinical candidate, or safe compound. They describe what computational work is justified by the evidence from this run.

15. Provenance and verification instructions

The reproducible package accompanying this dossier includes a `manifest.json` with the role, size, and SHA-256 of each file, and a sorted `checksums.sha256` file that detects missing, modified, or added files.

Unsealed provenance fields in this run: configuracion_sha256, vocabulario_especies.

How to verify the package without running anything: (1) open the ZIP and read `manifest.json`, which declares each file's role, size, and SHA-256, including missing files and their reason; (2) recalculate each file's SHA-256 and compare it with `checksums.sha256`; (3) check that no file is missing or extra relative to the manifest. The ZIP itself, `checksums.sha256`, and the `manifest.json` entry inside `files` are excluded by design—a file cannot contain its own hash—and the package README explains this.

A correct checksum demonstrates INTEGRITY: the bytes have not changed since the package was created. It does not show that the method is suitable, the preparation was correct, or the conclusion is true. Integrity is not scientific validity, and this dossier does not present them as the same.

These hashes let a third party verify that this dossier's conclusions were calculated from the inputs it declares. An on-chain record certifies WHEN the document was issued; it does not certify WHAT it was calculated from. Both are necessary; neither replaces the other.

16. Status glossary

Status	Meaning
NOT DEFINED	The case did not declare this context.
NOT EVALUATED	The control was not evaluated in this run.
NOT AVAILABLE	The expected result or artifact is unavailable.
NOT APPLICABLE	The control does not apply to this case.
REVIEW	Evidence is available, but it does not support a conclusion on its own.
ABSTENTION	Required evidence is missing or a declared blocker exists.
RECORDED	The data exists and was serialized by the run.
PASSED	The control ran and passed.

Appendix A. Legacy outputs (not for decisions)

The following indices are calculated by the pipeline for compatibility with earlier reports. **They are not decision criteria.** They are not probability, confidence, or quality measures, and should not be used to compare molecules or prioritize experimental work.

Field	Content	Status
Legacy composite index (total_score)	64.5 Without formula or published version: it is not interpretable as probability, confidence or quality.	REVIEW
Legacy affinity index	89.6 Without formula or published version: it is not interpretable as probability, confidence or quality.	REVIEW
Legacy ADME index	100.0 Without formula or published version: it is not interpretable as probability, confidence or quality.	REVIEW
Legacy drug-likeness index	76.9 Without formula or published version: it is not interpretable as probability, confidence or quality.	REVIEW